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PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:	**TechPort:**		No
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Zip Code:	21224-6823	**Congressional District:**	3
Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Hursh, Steven (Institutes for Behavior Resources, Inc.)		
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The aims of this project focus upon the development of an experimental test bed for modeling performance effectiveness and psychosocial adaptation in support of exploratory spaceflight missions beyond Earth's atmosphere. The research methodology employed provides for distributed interactive multi-person simulations in computer-generated environments focused upon psychosocial performance interactions within and between space-dwelling flight crews and Earth-based control center groups. The simulation provides an automated means for setting the context for the analysis of space crew performance as well as monitoring electronically the interactive effects of simulated communication modality constraints, mission management systems, and other stressful conditions. Within this context, the objectives of this project are to provide risk assessment and countermeasure evaluation of the following fundamental behavioral interaction operations that will most likely affect crew performance effectiveness and psychosocial adaptation: 1) the structure and function of communication channels within and between simulated space-dwelling and Earth-based groups; 2) factors associated with variations in the behavioral management systems between space-dwelling and

Task Description:	Earth-based groups; 3) factors associated with variations in workloads, stressful time pressure, and conflict conditions; and 4) behavioral and psychosocial interaction systems between spaceflight crews and Earth-centered mission support operations that are most likely to influence individual and group performance during long-duration missions. The results of these distributed interactive simulation studies have shown clearly that cooperative and productive interactions are maintained between individually isolated and dispersed crewmembers in a task-driven environment, and that experimental flight crews actively engage in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, investigations of communication modality constraints have shown a high degree of interchangeability between available communication modes that acted as a countermeasure to maintain effective crew performance. Further, variations in crew configurations have shown that performance effectiveness levels are significantly reduced during missions in which crew assignment changes resulted in inexperienced crew configurations, as contrasted with highly-experienced crew configurations. Studies of positive and negative incentive conditions on crew performance effectiveness have also shown increased performance effectiveness under both positive and negative incentive conditions, but an adverse change in psychosocial adaptation. Thus incentive conditions associated with simulated spaceflight missions can significantly affect psychosocial adaptation without compromising task performance effectiveness in trained and experienced volunteer crews. Newly-initiated studies on management system variations consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Tests of these methods revealed clear benefits of autonomy that improved crew performance and decreased negative psychosocial adaptation self-report measures. Autonomous missions also decreased undirected linguistic expressions of negative emotion but increased expressions of social processes and achievement. In addition, salivary cortisol and alpha-amylase (physiological stress markers) were lower during autonomous versus scheduled missions. The psychosocial and performance effects of autonomy were generally robust to communications constraints despite a physiological stress reaction to the unexpected loss of within-crew audio and text-messaging capabilities. The newly-developed Team Performance Task (TPT) has moved from development into its application phase as a simple, rapid, and objective assay of the fundamental behavioral elements of group cohesion. Extensive laboratory tests have established the basic parameters for TPT application to intact work-groups, and the TPT shows remarkable sensitivity to external incentives, between-crew differences, and individual crew member affinity for pro-social behavior. Additionally, new investigations of crew autonomy on performance and psychosocial adaptation in longer-duration, 12-hour mission simulations have been conducted during different parts of a 24-hour circadian cycle (either 'Early': 9 am - 9 pm, or 'Late': 9 pm - 9 am). As with previous work, crew performance was higher and physiological stress reactivity lower under autonomous mission management conditions; however, 'Late' workloads produced significant decrements in individual performances that included reaction time deficits, attentional lapses, psychosocial communication reductions, and decreases in crew cohesion (as assessed by the TPT). Importantly, in spite of these numerous decrements, crew performance efficiency remained normal. Lapses in individual performances thus appeared to be compensated for in the overall crew performance, indicating the utility of multi-person crews as countermeasures to reductions in individual performances under adverse workload conditions. Nevertheless, the dramatic decrease in psychosocial communication during late missions could be an early warning sign of crew desynchronization that could hinder crew cohesion and adaptability to malfunctions, environmental disruptions, or schedule demands. Future plans involve continuing these extended mission management and workload studies, continued TPT development and application, and migration of the laboratory simulation software to an internet-based platform for application to operational settings and controlled analogue environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The methodological development associated with this NSBRI-funded research has provided a test bed for modeling performance effectiveness and psychosocial adaptation in computer-generated distributed interactive multi-person environments. This experimental setting provides an automated means for setting the task performance context as well as monitoring electronically the interactive effects of communication modality constraints and other environmentally stressful conditions. Such a distributed interactive experimental analysis can characterize the extent to which variations in the structure and function of communication channels within and between individuals and groups affect the ability to accurately discern and respond to the realities and requirements of both operational and interpersonal transactions. The effects of environmentally stressful time constraints, appetitive and aversive characteristics of programmed incentive conditions, workload conditions, and management system conditions can be analyzed experimentally in the context of such distributed interactive environments. Research conducted within the context of this distributed interactive simulation model can provide the basis for developing effective patterns of communication and problem solving strategies as well as a range of training procedures to enhance problem solving effectiveness. This project's research on better ways to assess, train and manage small team performance effectiveness under hazardous and stressful conditions is relevant to transportation agencies, military forces, and first responders. The Earth benefits to be derived from the research extend to small operational group selection and training procedures, to the management of stressful interactions, and to the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded knowledge base. In addition, the TPT has shown itself to be an effective and validated measure of behaviors necessary for effective team performance and productive social relationships, and thus may be of considerable value to the non-astronaut population. Specifically, in additional to its potential as a basic research tool, the TPT may be applicable to a wide range of organizations (e.g., the military, law enforcement agencies, emergency/first responder crews, medical/surgical teams, athletic teams, business organizations) where communication, coordination, and cooperation are critical for success. In addition, the TPT could be utilized in clinical settings as a standardized assessment of social function in populations with neuropsychiatric disorders characterized by social deficits (e.g., schizophrenia, Autism spectrum) to elucidate underlying mechanisms mediating socially-relevant symptoms, test the efficacy of potential treatments, and document individual improvement over time while under treatment.

Task Progress:	Task progress includes newly-initiated studies on the effects of management system variations and workload condition variations that consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Tests of these methods revealed clear benefits of autonomy resulting in improved performance and decreased negative psychosocial and biobehavioral adaptation measures. Studies during the current period have extended the experimental analyses described above to include variations in communication constraints under differing crew management conditions. Early results support the beneficial effects of autonomy, and suggest that these effects are robust to communication constraints. Taken together, these experiments reveal that crew autonomy during simulated spaceflight missions can positively affect team performance effectiveness and individual self-report measures of morale. Mission goals and behavioral health objectives can be satisfied, and even enhanced, under autonomous management conditions in well-trained and cohesive crews. Major software architecture changes were made to improve and expand features of the geologic simulation. Due to the older software's developmental and support limitations, the simulation has been rewritten in a more optimal, industry standard development environment (ActionScript, Adobe Flash) that has speeded and simplified development, and now affords more flexibility in simulation design, higher quality simulation production, greater ease of software change for future growth, and the capability to operate transparently over local networks or over the Internet. Major functional enhancements of the software include: separation of 'Mission Control' and 'Flight Operations' roles, increased numbers of vehicles participating in scenarios (i.e., increased team sizes), and the addition of actual lunar surface backgrounds. Crews have already successfully interacted with the new simulation software, including successful 3-person interactions via the internet from remote locations (e.g., individuals located separately at IBR, The Johns Hopkins University School of Medicine, and UMBC). Conceptual development of the Team Performance Task (TPT) has undergone several prototype designs and programming refinements, and data collection aimed at building an empirical foundation for defining the task's parameters has begun. Extensive parametric testing has been completed in groups with and without a shared work history. Experimental assessments of Individual vs. Collective reward schemes are near completion, and the TPT is now being applied to regular geologic simulation missions. So far, the 6-min TPT shows remarkable sensitivity to external incentives, between-crew differences, and individual crew member affinity for pro-social behavior.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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